

Lab 13

Permutational ANOVA and permutational MANOVA

There is a clever alternative to sums-of-squares based ANOVA that compares groups with distances measures, which does not require any assumptions about distributions. Conceptually, this is very similar to ANOVA. In ANOVA, the F-value is the ratio of variance between groups (signal) to variance within groups (noise). The bigger the signal to noise ratio, the larger the F-value, which can be converted to a p-value.

This also works with distances: Instead of an F-value, I can calculate a Delta-value (or an equivalent statistic that goes by some other name), which is the ratio of distances between observations between groups (signal) to distances within groups (noise). This gives me a signal to noise ratio that is my Delta value (equivalent to an F value). Next, I need to compare this to an expected distribution of that value (like an F-distribution).

The problem is that we don't know how that distribution looks like for non-normal data. So, the permutational ANOVA generates this distribution empirically, by randomly shuffling the class variables within the dataset. Now that the class variable has been randomized relative to the actual observations, you calculate a new randomly expected Delta-value. You randomize (permute) your class variable again, and calculate another Delta-value.

Repeat 10,000 times and you get a Delta distribution specific to your type of data that you can use to get the p-value to indicate significance between groups. This is simply the position (percentile) of your original Delta-value on the distribution of 10,000 values that you generated. Hence, you assume nothing about the shape of the distribution.

This gets a bit more complicated when you have to deal with interactions, nested treatments, covariates, random and fixed effects, and pairwise multiple comparisons. The FORTRAN program "Permanova6" by Marti Jane Anderson of the Department of Statistics, University of Auckland, is an excellent implementation that can do almost anything. The only drawback is that your experimental design needs to be balanced (i.e. the same number of observations for each treatment).

13.1. Permanova6

Download the two packages "normal ANOVA" and "permutational ANOVA", unzip the folders and run both types of analysis for our complete "Lentil" dataset. You will see that Permanova6 produces almost identical results to the regular ANOVA.

13.2. Permutational Multivariate Analysis of Variance

Because Permanova6 cannot handle unbalanced designs, I include another version in R below, that is designed to analyze community data and that can handle unbalanced data well. However, it works only for multivariate dependent variables and I did not have much luck replicating results from sums-of-squares based (M)ANOVA. It seems the procedure somewhat errs on the side of making a type I error especially if interactions are involved.

- Download the dataset Reclaim.csv from the course website. This data represents species counts on reclamation sites that received different soil and fertilizer treatments to enhance the revegetation process. Import and check if the data imported correctly as usual:

```
rec=read.csv("reclaim.csv")
fix(rec)
```

- Next, we have to break the dataset up into separate files for treatments, which must be defined as a “factor” or class variable, and species frequency measurements, that must be defined as a matrix that contains only numbers:

```
soils=as.factor(rec[,1])
fertilizer=as.factor(rec[,2])
measurements=as.matrix(rec[,3:32])
```

- If you look at histograms of the species data, you quickly recognize that you could not possibly use an ANOVA or MANOVA to analyze these data. They are non-normal and no transformation can change that. Check two or more of the species:

```
hist(rec$Achmil)
hist(rec$Empnig)
```

- Permutational Multivariate Analysis of Variance is included in the “vegan” package, which you will have to install. You have to choose a distance measure. For regular measurements, that would usually be “euclidean”, but for species community data, “bray” (the Bray-Curtis distance) is more appropriate. The following will give you an output exactly equivalent to an ANOVA table:

```
library(vegan)
adonis(measurements~soils, distance="bray")
adonis(measurements~fertilizer, distance="bray")
adonis(measurements~fertilizer*soils, distance="bray")
adonis(measurements~fertilizer+soils, distance="bray")
```

- These statistics tell you whether your soil conditions and fertilizer treatments significantly affect your community composition (i.e. you have a multivariate dependent variable). If you want to know how they are influenced for individual species, you need to follow this up with an ordination, which we’ll cover in the next lab.